Do Now · retrieval

Lesson 072 · Algorithm Design: From Specification to Steps

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is a loop invariant?

Q2

What is a postcondition?

Q3

How is an input specification useful?

Q4

C# array a=[3,5,7]: set a[1]=a[0]+a[2]. Give final array and sum.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Observe — Read the problem to identify exact input type, required output, boundaries and invalid cases.

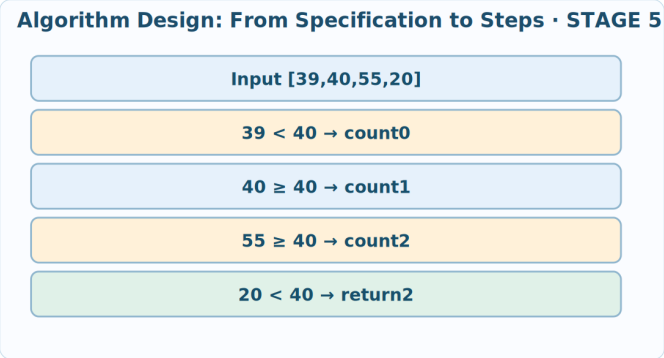
Core idea 2

Explain — Write an algorithm with an invariant: after each iteration, count is the number of qualifying values examined so far.

Core idea 3

Evaluate — Choose meaningful tests around the 40 boundary and compare output against an independently calculated result.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

algorithm — Finite unambiguous sequence of steps.

invariant — Property preserved across loop iterations.

precondition — Required input assumption.

postcondition — Expected condition after executing.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

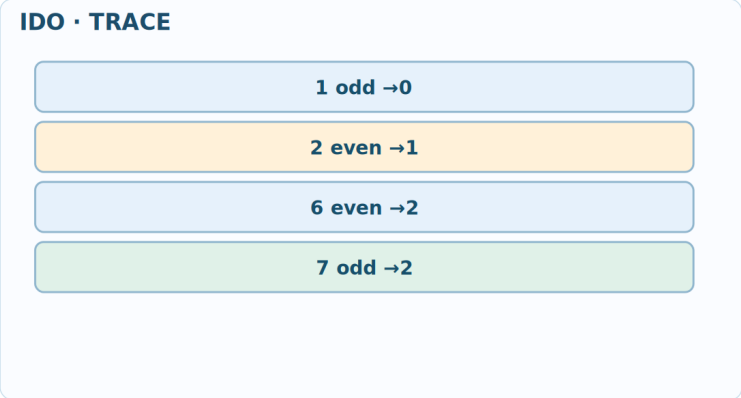
Problem

Design a function counting even numbers in [1,2,6,7].

Follow the worked method

1. Explain — Using modulo2==0 identifies even values; the list has 2 and 6.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Explain
Zero does not count as negative, so use <0, not <=0.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Trace CountPasses for [40,41,39,100]. Show count after each item. [6]

Question 2

Write C# method that counts negative integers in an array, and returns0 for empty. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain a loop invariant, time complexity and edge tests for CountPasses. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

An algorithm can be evaluated without considering its input requirements.

Use key vocabulary from memory

1. Explain algorithm in your own words.

2. Explain invariant in your own words.

3. Explain precondition in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.